

231501137

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Exercise 3: Google App Engine SDK and Web App Development

Aim

To install the Google App Engine SDK and develop a Hello World / simple web application using Python.

Procedure

Step 1: Install Google Cloud CLI & Python

Ensure Python 3 is installed on the system.

```
ubuntu@ubuntu:~$ sudo apt install python3
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 3
```

Download and install the Google Cloud CLI (contains the modern App Engine SDK) from cloud.google.com/sdk/docs/install:

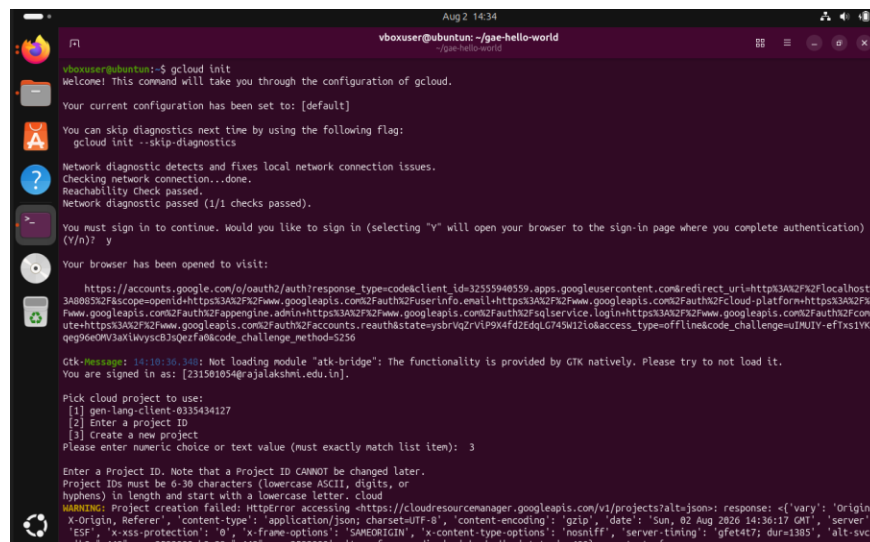
```
curl -O https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-linux-x86_64.tar.gz
```

```
tar -xvf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
```

Step 2: Initialize GCloud & Install Components

gcloud init

Sign in to your Google account when prompted.



```
Aug 2 14:34
vboxuser@ubuntu: ~/gae-hello-world
vboxuser@ubuntu:~$ gcloud init
Welcome! This command will take you through the configuration of gcloud.

Your current configuration has been set to: [default]

You can skip diagnostics next time by using the following flag:
gcloud init --skip-diagnostics

Network diagnostic detects and fixes local network connection issues.
Checking network connection...done.
Reachability check passed.
Network diagnostic passed (1/1 checks passed).

You must sign in to continue. Would you like to sign in (selecting "Y" will open your browser to the sign-in page where you complete authentication)
(Y/n): y

Your browser has been opened to visit:
https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Flocalhost%3A8080%2F&scope=openid%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fappengine.admin%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute%2Fhttps%3A%2F%2Fwww.googleapis.com%2Fauth%2Faccounts.reauth%2Fstate=ysbrvqzrVLP9X4fdEdqLG745K12Io&access_type=offline&code_challenge=UJUIV-efTxiYkyqeg96e0M73akWyscbJ3Qe2fAb&code_challenge_method=S256

Gtk-Message: 14:10:36.340: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
You are signed in as: [231501054@rajalakshmi.edu.in].

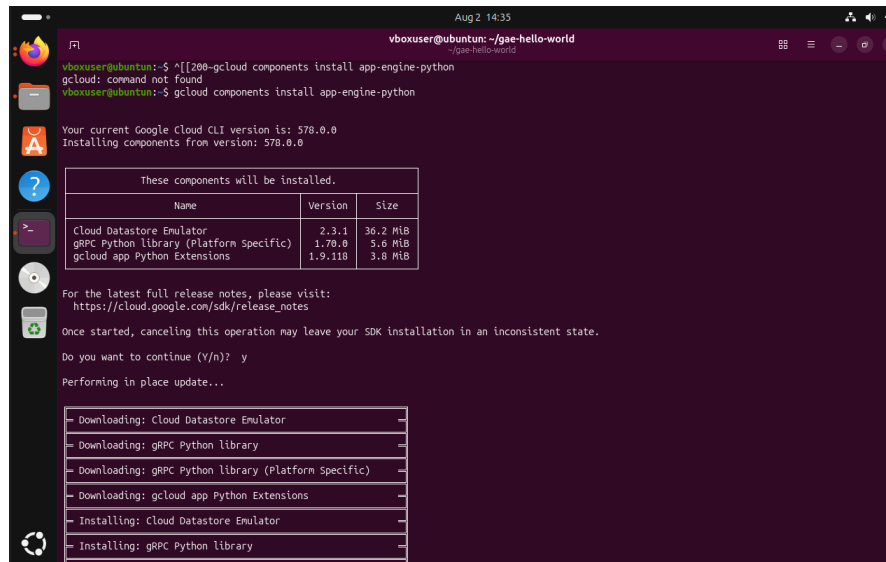
Pick cloud project to use:
[1] gen-lang-client-0335434127
[2] Enter a project ID
[3] create a new project
Please enter numeric choice or text value (must exactly match list item): 3

Enter a Project ID. Note that a Project ID CANNOT be changed later.
Project IDs must be 6-30 characters (lowercase ASCII, digits, or hyphens) in length and start with a lowercase letter. cloud
WARNING: Project creation failed: HttpError accessing <https://cloudresourcemanager.googleapis.com/v1/projects?alt=json>: response: <{'vary': 'Origin, X-Origin, Referer', 'content-type': 'application/json; charset=UTF-8', 'content-encoding': 'gzip', 'date': 'Sun, 02 Aug 2020 14:36:17 GMT', 'server': 'ESF', 'x-xss-protection': '0', 'x-frame-options': 'SAMEORIGIN', 'x-content-type-options': 'nosniff', 'server-timing': 'gfe47; dur=1305', 'alt-svc': '<h3>=443'; ma=2592000 h3-20='<h3>=443'; ma=2592000'}>, transfer-encoding: chunked, 'status': 400}>, content: <f
```

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gcloud components install app-engine-python



The screenshot shows a terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' with the date 'Aug 2 14:35'. The user enters the command `gcloud components install app-engine-python`. The terminal output shows the current Google Cloud CLI version is 578.0.0 and that components are being installed from version 578.0.0. A table lists the components to be installed: Cloud Datastore Emulator (2.3.1, 36.2 MiB), gRPC Python library (Platform Specific) (1.70.0, 5.6 MiB), and gcloud app Python Extensions (1.9.118, 3.8 MiB). Below the table, a link to release notes is provided. The user confirms the installation with 'y', and the terminal shows the progress of downloading and installing the components.

```
vboxuser@ubuntu:~/gae-hello-world$ gcloud components install app-engine-python
gcloud: command not found
vboxuser@ubuntu:~/gae-hello-world$ gcloud components install app-engine-python

Your current Google Cloud CLI version is: 578.0.0
Installing components from version: 578.0.0

These components will be installed.


| Name                                    | Version | Size     |
|-----------------------------------------|---------|----------|
| Cloud Datastore Emulator                | 2.3.1   | 36.2 MiB |
| gRPC Python library (Platform Specific) | 1.70.0  | 5.6 MiB  |
| gcloud app Python Extensions            | 1.9.118 | 3.8 MiB  |



For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, canceling this operation may leave your SDK installation in an inconsistent state.

Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
```

Step 3: Create the App Directory

```
mkdir gae-hello-world
```

```
cd gae-hello-world
```

Step 4: Create Application Code (main.py)

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def hello():
```

```
    return "Hello World from Google App Engine!"
```

```
if __name__ == "__main__":
```

```
    app.run(host="127.0.0.1", port=8080, debug=True)
```

Step 5: Create Configuration File (app.yaml)

```
runtime: python39
```

```
handlers:
```

```
- url: /*
```

```
  script: auto
```

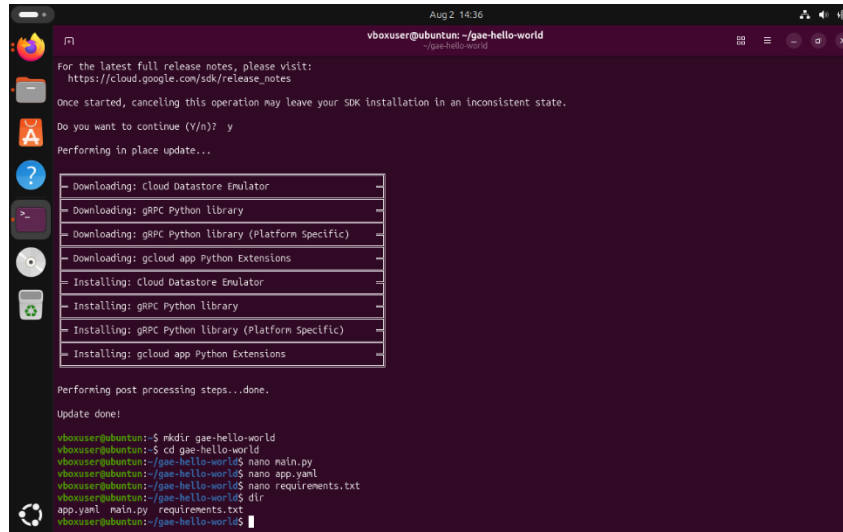
Step 6: Create requirements.txt

```
Flask==2.0.1
```

Output

Files created in project directory (main.py, app.yaml, requirements.txt):

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A terminal window titled 'vboxuser@ubuntu: ~/gae-hello-world' showing the installation of the Google App Engine SDK. The terminal displays instructions to visit the release notes, a confirmation to continue, and a progress bar for downloading and installing various components. The components listed are: Cloud Datastore Emulator, gRPC Python library, gRPC Python library (Platform Specific), and gcloud app Python Extensions. The installation is shown as complete, followed by post-processing steps and a directory listing of the installed files.

```
Aug 2 14:36
vboxuser@ubuntu: ~/gae-hello-world

For the latest full release notes, please visit:
https://cloud.google.com/sdk/release_notes

Once started, cancelling this operation may leave your SDK installation in an inconsistent state.
Do you want to continue (Y/n)? y
Performing in place update...

- Downloading: Cloud Datastore Emulator
- Downloading: gRPC Python library
- Downloading: gRPC Python library (Platform Specific)
- Downloading: gcloud app Python Extensions
- Installing: Cloud Datastore Emulator
- Installing: gRPC Python library
- Installing: gRPC Python library (Platform Specific)
- Installing: gcloud app Python Extensions

Performing post processing steps...done.
Update done!

vboxuser@ubuntu: $ mkdir gae-hello-world
vboxuser@ubuntu: $ cd gae-hello-world
vboxuser@ubuntu: ~/gae-hello-world$ nano main.py
vboxuser@ubuntu: ~/gae-hello-world$ nano app.yaml
vboxuser@ubuntu: ~/gae-hello-world$ nano requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$ dir
app.yaml  main.py  requirements.txt
vboxuser@ubuntu: ~/gae-hello-world$
```

Result

Thus, the Google App Engine SDK was installed and a simple Hello World web application was developed using Python and Flask.